

AWS Cloud Technical Essentials

Introduction to AWS Identity and Access Management (AWS IAM)

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Objectives

After completing this lab, you will be able to:

- Explore IAM Users and Groups
- Inspect IAM policies applied to groups
- Follow a real-world scenario that adds users to groups and explores group permissions
- Locate and use the IAM sign-in URL
- Experiment with policies and service access

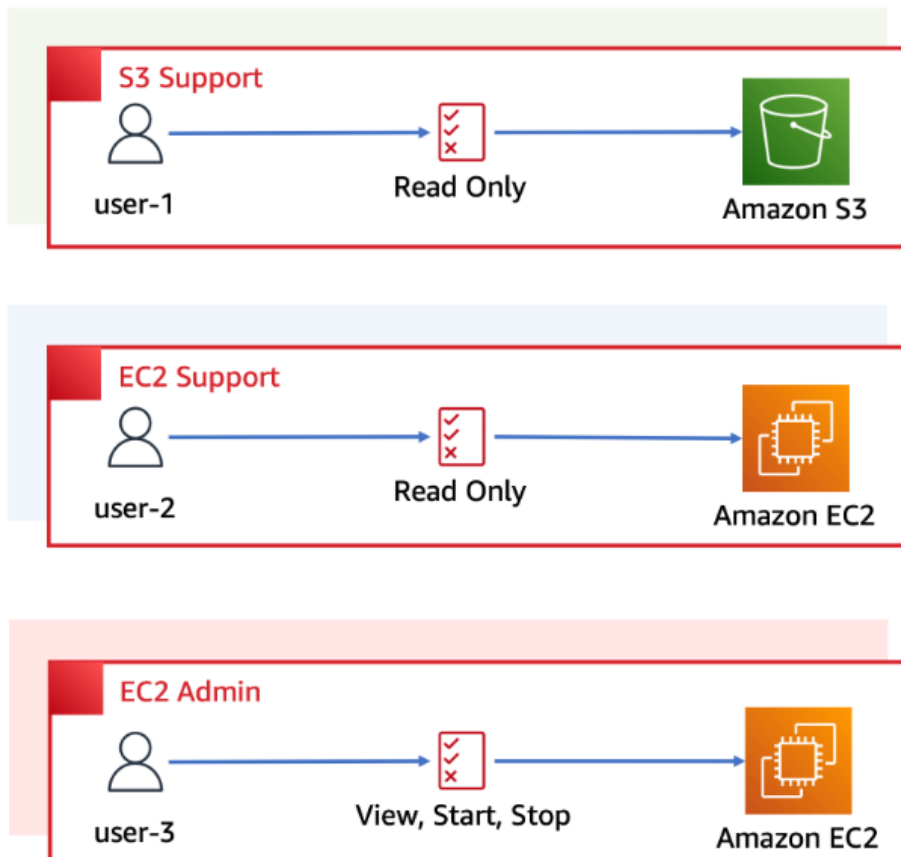
Lab Overview

This lab provisions the following IAM resources for you to explore:

Resource	Details
Three users	user-1, user-2, and user-3
Three groups	S3 Support, EC2 Support, and EC2 Admin
S3 Support	Read-Only access to Amazon S3
EC2 Support	Read-Only access to Amazon EC2
EC2 Admin	View, Start, and Stop EC2 instances

IAM access model

The diagram below summarizes the user-to-group-to-service permissions used throughout the lab.



Important: You might receive error messages when performing actions beyond the steps provided in this lab guide. The lab relies on IAM, which limits your access to the services authorized for use in the lab. These error messages will not impact your ability to complete the lab.

Start lab

1. To launch the lab, at the top of the page, choose Start Lab.

Caution: You must wait for the provisioned AWS services to be ready before you can continue.

2. To open the lab, choose Open Console.

You are automatically signed in to the AWS Management Console in a new web browser tab.

Warning: Do not change the Region unless instructed.

Common sign-in errors

Error: Choosing Start Lab has no effect

In some cases, certain pop-up or script blocker web browser extensions might prevent the Start Lab button from working as intended. If you experience an issue starting the lab:

- Add the lab domain name to your pop-up or script blocker's allow list or turn it off.
- Refresh the page and try again.

Task 1.1: Explore IAM users, groups, and policies

In this task, you explore the IAM users and groups that were created for you.

3. At the top of the AWS Management Console, in the search bar, search for and choose IAM.

4. In the left navigation pane, choose Users.

The list contains the existing users: user-1, user-2, and user-3.

5. Choose the user-1 link.

6. Choose the Groups tab.

7. Choose the Security credentials tab.

The summary page for user-1 opens with the Permissions tab displayed. Notice that user-1 does not have any permissions.

Note: user-1 does not belong to any groups.

In the Console sign-In section, note that user-1's Console password is enabled.

8. In the left navigation pane, choose User groups.

9. Choose the EC2-Support group name link.

10. In the summary page for the EC2-Support group, choose the Permissions tab.

This group has a managed policy associated with it — AmazonEC2ReadOnlyAccess. Managed policies are pre-built policies (built either by AWS or by your administrators) that can be attached to IAM users and groups. When updated, all users and groups associated with the policy inherit the changes to that policy.

11. Expand the Plus icon to expand the contents of AmazonEC2ReadOnlyAccess.

A policy defines the actions allowed or denied for specific AWS resources. This policy grants permission to list and describe information about EC2, Elastic Load Balancing, Amazon CloudWatch, and Auto Scaling. Support role policies commonly restrict access to a read-only level, without the ability to modify AWS resources.

Basic structure of an IAM policy:

- Effect indicates whether to Allow or Deny the permissions.
- Action specifies the API calls allowed to an AWS service, such as cloudwatch:ListMetrics.
- Resource defines the scope of entities covered by the policy rule, such as a specific S3 bucket or EC2 instance, or * for any resource.

12. In the left navigation pane, choose User groups.

13. Choose the S3-Support group name link.

14. Choose the Permissions tab.

15. Expand the Plus icon to expand the contents of AmazonS3ReadOnlyPolicy.

The S3-Support group has the AmazonS3ReadOnlyAccess policy attached. This policy has permissions to get and list resources from Amazon S3.

16. In the left navigation pane, choose User groups.

17. Choose the EC2-Admin group name link.

This group differs slightly from the other two groups. Instead of a managed policy, it has an inline policy, a policy assigned to one discrete user or group, and typically used to apply permissions for one-off situations.

18. Choose the Permissions tab.

19. Expand the Plus icon to expand the contents of EC2-Admin-Policy.

This policy grants permission to view (describe) information about Amazon EC2 resources, start and stop instances, and describe CloudWatch alarms.

Task 1.2: Manage users and groups

In this task, you work with the users and groups, enabling permissions to support a business scenario.

You can ignore any “not authorized” errors that appear during this task caused by limited permissions in your lab account. The error messages will not impact your ability to complete the lab.

Business scenario

Your company’s use of Amazon Web Services grows over time. The company uses many Amazon EC2 instances and a significant amount of Amazon S3 storage. You want to grant access to new staff members based on their job functions, as shown in the table.

User	Group	Permissions
user-1	S3-Support	Read-Only access to Amazon S3
user-2	EC2-Support	Read-Only access to Amazon EC2
user-3	EC2-Admin	View, Start, and Stop Amazon EC2 instances

Add user-1 to the S3-Support group

You recently hired user-1 into a role to provide support for Amazon S3. You want to add them to the S3-Support group to inherit the necessary permissions in the attached AmazonS3ReadOnlyAccess policy.

20. In the left navigation pane, choose User groups.
21. Choose the S3-Support group name link.
22. In the Users tab, choose Add users.
23. Select user-1.
24. At the bottom of the screen, choose Add users.

Expected output: Users added to this group. In the Users tab, under the Users in this group section, the list should now include user-1.

Add user-2 to the EC2-Support group

You hired user-2 for a role to provide support for Amazon EC2.

25. Repeat the steps you used for user-1 to add user-2 to the EC2-Support group.
26. Confirm user-2 is part of the EC2-Support group.

Add user-3 to the EC2-Admin group

You hired user-3 as your Amazon EC2 administrator.

27. Repeat the steps you used for user-1 to add user-3 to the EC2-Admin group.
28. Confirm user-3 is part of the EC2-Admin group.

Review user and group assignments

29. In the left navigation pane, choose User groups.

Each group should have a 1 in the Users column to denote the number of users in each group.

If you do not have a 1 beside each group, revisit the preceding instructions to ensure that each user belongs to a group, as shown in the table in the Business Scenario section.

Congratulations! You have successfully explored and managed IAM users, groups, and policies.

Task 2: Use the IAM sign-in URL

In this task, you test the permissions of each IAM user.

Task 2.1: Locate and access the IAM sign-in URL

30. In the left navigation pane, choose Dashboard.

In the AWS Account section, on the right side of the dashboard, locate the Sign-in URL for IAM users in this account URL. The URL format appears as follows:

```
https://123456789012.signin.aws.amazon.com/console
```

31. Use this link to sign in to the AWS account you are currently using.

32. Copy the IAM users sign-in link to a text editor.

You need this link to sign in later, testing each user.

Task 2.2: Log in with different IAM users

33. In your browser, open a private window.

If you are using Firefox:

- In the upper-right corner, choose the Open application menu icon and choose New Private Window.

If you are using Chrome:

- In the upper-right corner, choose the Customize and Control Google Chrome ellipsis icon and choose New incognito window.

If you are using Edge:

- In the upper-right corner, choose the Settings and more ellipsis icon and choose New InPrivate window.

If you are using Internet Explorer:

- On the Tools menu, choose InPrivate Browsing.

34. Paste the IAM users sign-in link you copied earlier into the private window, and press Enter.

You will now sign in as user-1, your new Amazon S3 storage support person.

Sign in as user-1

35. In the sign-in screen, use the following information:

- IAM user name: user-1
- Password: Paste the value of UserPassword located to the left of these instructions.
- Press Enter.

36. At the top of the AWS Management Console, in the search bar, search for and choose S3.

37. From the S3 dashboard, choose the link for the bucket that has s3bucket in its name.

To the left of these instructions, you can also find the name of your S3 bucket. A bucket with awslabs-resources in its name might also be present, along with an error. This is normal. You do not have access to this bucket.

Because your user is part of the S3-Support group in IAM, the user has permission to view a list of Amazon S3 buckets and the contents of the s3bucket.

Next, test whether user-1 has access to Amazon EC2.

38. Search for and choose EC2.

39. Locate your current region in the upper right side of the EC2 console. Verify this region is the same as the Region value in the left side pane of these instructions.

If not, choose the drop-down arrow at the top of the screen, to the left of user-1, and select the Region value that matches the Region to the left of the instructions.

40. In the left navigation pane, in the Instances section, choose Instances.

You should receive an error message: You are not authorized to perform this operation. You cannot see any instances because your user has not been assigned any permissions to use Amazon EC2.

Sign in as user-2

Next, you sign in as user-2, your Amazon EC2 support person.

41. In the AWS Management Console, sign out user-1: choose user-1, then choose Sign out.

42. Paste the IAM users sign-in link into your private window and press Enter.

43. Sign in with:

- IAM user name: user-2
- Password: Paste the value of UserPassword located to the left of these instructions.
- Press Enter.

44. Search for and choose EC2.

45. Navigate to the Region where your lab launched, if you are not currently in that Region.

Choose the drop-down arrow at the top of the screen, to the left of user-2, and select the Region value that matches the Region to the left of the instructions.

46. In the left navigation pane, in the Instances section, choose Instances.

You can view an Amazon EC2 instance because you have read-only permissions. However, you cannot make any changes to Amazon EC2 resources. You should see a single running EC2 instance.

47. Make sure that the EC2 instance is selected.

48. On the Instance state menu, choose Stop instance.

49. In the Stop instance? window, choose Stop.

Note: You should receive an error stating that You are not authorized to perform this operation. This demonstrates that the policy only allows you to view the information, without making changes.

50. Close the error message by clicking the X in the top of the red error window.

Next, check to see if user-2 can access Amazon S3.

51. Search for and choose S3.

The S3 console should return an empty list of buckets because user-2 does not have permission to use Amazon S3. The Copy ARN, Empty, Delete, and Create options provide no access to those features.

Sign in as user-3

52. Sign in as user-3, your Amazon EC2 administrator.

53. Sign out user-2 by choosing user-2, then Sign out.

54. Paste the IAM users sign-in link into your private window and press Enter.

55. Sign in with:

- IAM user name: user-3
- Password: Paste the value of UserPassword located to the left of these instructions.
- Press Enter.

56. Search for and choose EC2.

57. Navigate to the Region where your lab launched, if you are not currently in that Region.

Choose the drop-down arrow at the top of the screen, to the left of user-3, and select the Region value that matches the Region to the left of the instructions.

58. In the left navigation pane, in the Instances section, choose Instances.

As an EC2 administrator, you should have permission to stop the Amazon EC2 instance.

59. Make sure that the EC2 instance is selected.

60. On the Instance state menu, choose Stop instance.

61. In the Stop instance? window, choose Stop.

The instance will enter the stopping state and shut down.

62. Close your private window.

Congratulations! You have successfully used the IAM sign-in URL, logged in with different users, and tested the access level of each user. Your newly hired employees are ready to start!

Lab complete

Congratulations! You can now:

- Explore IAM Users and Groups
- Inspect IAM policies applied to groups
- Follow a real-world scenario that adds users to groups and explores group permissions
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End lab

Follow these steps to close the console and end your lab.

63. Return to the AWS Management Console.

64. At the upper-right corner of the page, choose AWSLabsUser, and then choose Sign out.

65. Choose End Lab and then confirm that you want to end your lab.